

NovaTech Revenue Intelligence Dashboard

Executive Report

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1. Overview

This report summarizes the design, findings, and validation of the Revenue Intelligence Dashboard built to unify NovaTech's CRM, marketing, and support data. The goal was to replace the weekly manual report-pulling process with a single, self-serve view of the customer journey — from marketing touchpoint to closed deal to support experience — plus a natural-language chat interface (Amazon Quick) for ad-hoc questions.

2. Data Strategy

Three source systems were verified against the data dictionary, imported into SPICE (Amazon Quick's in-memory storage engine, which keeps data ready for fast querying), and prepared for analysis:

- CRM Deals (499 deals): we added a days-to-close field and a won/lost flag, and corrected a date-format error that had been silently blanking out every deal date.
- Marketing Campaigns (2,240 leads): we added a campaign return-on-investment field and a total product spend field, and removed consumer-demographic fields that do not apply to a business-to-business audience.
- Support Tickets (3,000 tickets): we added a resolution-time field and an unresolved-ticket flag.

A fourth, unified dataset joins all three sources on the shared account identifier. It is anchored on CRM using a left join, a join type that keeps every row from the CRM table (all 85 of NovaTech's tracked accounts) even when a matching marketing or support record does not exist. Because this join creates one row for every combination of deal, lead, and ticket belonging to the same account, any total built on the unified dataset would otherwise overcount unless it is calculated per unique account, deal, or ticket rather than per row — this distinction matters and is explained further in Section 5.

3. Dashboard Design

Marketing Funnel

This sheet shows campaign performance by channel, progression through the funnel stages, campaign return on investment, and response rates. It includes filter controls for campaign, channel, date range, and customer segment.

Sales Pipeline

This sheet shows deal stages and loss reasons, revenue by segment and product, win rates by region, sales representative, and product, and a trend of days-to-close over time. It includes filter controls for region, sales manager, product, date range, and deal stage, and one-click filtering is enabled on two visuals so a user can click a bar to filter the rest of the sheet.

Customer Health

This sheet shows ticket volume by account, resolution time by priority level, product-area issues, and sentiment distribution. It also includes an account-risk scatter plot, the required visual built from the unified dataset, which plots ticket volume against deal value, colors each point by customer sentiment, and is filtered to the twenty accounts with the highest ticket volume. A navigation action on this sheet lets a user click through to the Sales Pipeline sheet for account-level follow-up.

4. Natural-Language Access: Topic Configuration

Quick Chat is Amazon Quick's natural-language question-and-answer tool, and a Topic is a configuration layer that tells Quick Chat what business terms mean so it can answer questions more accurately. A Quick Chat baseline was recorded by asking three representative questions before any Topic configuration. A Topic ("NovaTech Revenue Intelligence Topic") was then built on the unified dataset with business-friendly field names, plain-language synonyms, two new calculated fields (Conversion Rate and Win Rate), and internal identifier fields hidden from view.

Effect on accuracy: single-value questions (overall averages, simple rankings) were answered correctly both before and after Topic setup. Cross-dataset questions that failed before configuration — such as win rate by campaign source — became answerable after the Topic linked campaign and deal fields together. However, Topic setup also surfaced a new, recurring failure mode: for any question requiring a summed total on the unified dataset (total revenue, total campaign ROI), Quick Chat summed the underlying ratio or dollar value across every duplicated join row instead of computing one true value per deal or campaign, producing totals inflated by roughly 10–25×. This is the same row-multiplication issue the dashboard's calculated fields were specifically built to avoid — Quick Chat's on-the-fly aggregation does not consistently apply the same safeguard.

Two smaller accuracy gaps were also found and are easy to fix with more specific synonyms: Quick Chat twice substituted a similarly-named field (Contact Channel for Product Area; Employee Count for Company Size Tier) when asked a question whose exact field name had overlapping synonyms in the Topic.

5. Key Insights and Recommended Actions

5.1 — Marketing funnel bottleneck at Qualified Lead

Finding: Lead-to-Qualified-Lead conversion drops 82% (434 → 78 leads), the steepest decline of any funnel stage.

Why it matters: This is where the majority of marketing-generated leads are lost before ever reaching a sales rep, limiting how much campaign spend converts into pipeline.

Recommended action: Review lead-qualification criteria and the marketing-to-sales handoff process to identify why leads are being disqualified at this stage.

5.2 — All six campaigns are currently unprofitable

Finding: Every active campaign shows negative ROI (roughly -0.90 to -1.00), meaning spend is not being recovered through attributed revenue on any channel.

Why it matters: NovaTech is not currently seeing a positive return on its ~\$90K in total campaign spend across any channel.

Recommended action: Reallocate budget away from the lowest-ROI campaign (NovaEdge Awareness) toward relatively stronger performers pending a full attribution review.

5.3 — Two distinct causes behind nearly half of lost deals

Finding: "Poor Product Fit" and "No Decision Made" are tied as the top loss reasons (43 deals each), together accounting for nearly half of all 184 lost deals.

Why it matters: These point to two different root causes — product-market mismatch versus a sales-process/urgency gap — and each needs a distinct fix.

Recommended action: Sales enablement should audit "No Decision Made" notes for urgency-building opportunities, while product marketing investigates "Poor Product Fit" losses for positioning or targeting issues.

5.4 — Two accounts show the clearest churn-risk signal

Finding: Account ACCT-045 has the highest average deal value (~\$9,500) among negative-sentiment, high-ticket-volume accounts; ACCT-041 has the highest ticket volume (~68 tickets) among the same group.

Why it matters: These are exactly the high-value, high-friction, negative-sentiment accounts most likely to churn.

Recommended action: Assign a dedicated customer success manager to proactively contact ACCT-045 and ACCT-041 within two weeks, ahead of renewal risk increasing further.

5.5 — Critical tickets are barely faster than high-priority tickets

Finding: “High” priority tickets resolve in roughly the same time as “Critical” tickets (~60 vs. ~55 hours) — a gap of under 5 hours despite the urgency difference.

Why it matters: If critical issues are not being meaningfully fast-tracked in practice, the most severe customer problems may be experiencing longer-than-expected waits, directly affecting sentiment and retention.

Recommended action: Review the support team's triage and SLA process to confirm critical tickets are properly escalated ahead of high-priority ones.

6. Where AI Analysis Agreed or Disagreed With the Dashboard

Five representative questions were asked of Quick Chat after Topic configuration and cross-checked against the published dashboard:

Question Asked	Result	Notes
Which campaign channel has the highest conversion rate?	Partial	Q correctly ranked Partner Referral first, but its totals were inflated into the thousands because it summed a per-deal ratio across duplicated rows instead of calculating one rate per campaign.
Average deal size by company size?	No	Q's overall average matched the dashboard exactly, but Q grouped results by the raw employee count instead of the intended Small, Medium, Large, and Enterprise size categories.
Resolution time: critical vs. low priority?	No	Q's overall average matched the dashboard, but Q grouped results by date instead of recognizing that “critical” and “low” refer to priority levels.
Top 10 accounts by tickets and total deal revenue?	No	The ticket count Q reported was accurate, but the deal revenue figure was inflated by roughly twenty-five times due to the same row-duplication issue described in Section 2.
Any campaigns spending more than they earned back?	Partial	Q correctly identified all six campaigns as unprofitable and correctly ranked which channel lost the least money, but the reported dollar magnitude of the loss was inflated.

Bottom line: Quick Chat is reliable for directional questions, rankings, and simple single-value lookups. For any total, sum, or precise dollar figure on the unified dataset, the published dashboard should be treated as the source of truth — its calculated fields already apply the distinct-count and per-entity averaging techniques that Quick Chat's on-the-fly aggregation does not consistently replicate.

7. Next Steps

- We recommend adding explicit synonyms, such as “company size” and the specific priority level names, to reduce the two field-ambiguity gaps found during Quick Chat testing.
- We recommend investigating whether a pre-aggregated summary table can prevent the inflated totals seen in Quick Chat's answers to questions that require summing values across the unified dataset.
- We recommend sharing the five key insights above with the revenue team at Monday's meeting, alongside the recommended actions listed in Section 5.